

Learn values & states in an env.

Professor draws 2x3 maze to give an example

$$r(s) \forall s$$

* = optimal or near optimal

π_* optimal policy

perform episodes of training - looking at first-visit MC prediction, for est. $V \approx v_\pi$

large # of episodes

compute returns in reverse order

* looking for unique states only *

π : an eqi-random policy (say)

what is best action in each state

can we learn a better policy over time

make a small change, and see if we improve to π $q(s,a)$ values

Now looking at S.3 Monte Carlo ES for est: $\pi \approx \pi_*$

- evaluation

- improvement

$$\pi_0 \xrightarrow{\text{eval}} q(s,a) \xrightarrow{\text{improve}} \pi_1 \text{ repeat until } \rightarrow \pi_*$$

first state can be any state in our state space

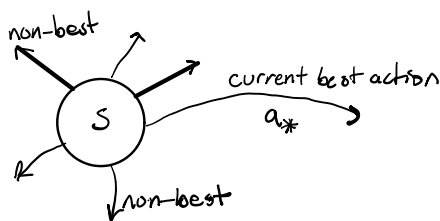
S.4 MC no ES (Claude insert algo here from book for on-policy below)

on-policy first visit MC control (for ϵ -soft π) est $\pi \approx \pi_*$

do best all the time = exploitation - best action in state all the time

not all the time = exploration - non best actions sometimes

Algo parameter small $\epsilon > 0$



let there be $A(s)$ actions possible in state s

$$A(s) = \{\uparrow, \downarrow, \leftarrow, \rightarrow, R\}$$

$$|A(s)| = \# \text{ of actions possible in state } s$$

Given: A small $\epsilon > 0$

every action is possible w/ @ least probability

$$\frac{\epsilon}{|A(s)|}$$

Epsilon Greedy algo

$$\text{e.g. } \epsilon = .10$$

$$\text{Every action is poss. w/ prob. @ least} = \frac{.10}{5} = .02$$

The prob assigned to a^* , the best action

$$= 1 - \epsilon + \frac{\epsilon}{|A(s)|}$$

$$= | -\epsilon + \frac{\epsilon}{|A(s)|}$$

5.5 off-policy prediction

On-policy learning: starts with policy π , improves repeatedly, towards π^* the best policy

off-policy learning:

Two policies:

- 1) target policy π
- 2) behavior policy B

Learn to estimate π^* by acting using policy B

Policy B



want to learn policy π

$A_{0,\pi}$

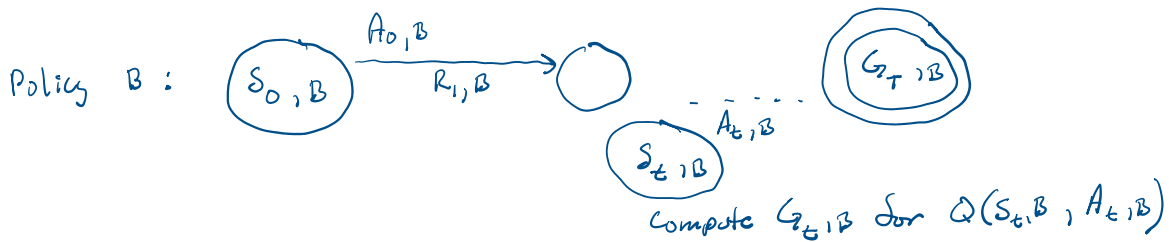
agent learns from an expert that already has knowledge
can the agent learn from the expert

off-policy has high variance

" " needs another env to learn in or an expert, can cost a lot \$\$\$

but off policy learning is general

on policy learning is where $B = \pi$



Policy π : $G_{t,B}$ from B gets inserted here (check with book here)
to obtain $G_{t,\pi}$